

Jackett Management UI & System Database

â€” Design Spec

Revision: 1 Last modified: 2026-06-06T00:00:00Z

Field	Value
Date	2026-04-27
Author	Brainstorm session (milos85vasic.2nd@gmail.com + Claude)
Status	Awaiting user review
Phase	D (Go-canonical Jackett surface + SQLite); Phase 2 = port merge search to Go
Supersedes	n/a (extends 2026-04-26-jackett-autoconfig-clean-rebuild-design.md)

1. Problem Statement

The Jackett auto-configuration shipped in commits 45591c2 â†’ fc1f009 (Layers 1-7) configures private-tracker indexers from .env credentials at startup. Verified live behavior on 2026-04-27:

- 4 credential triples discovered (IPTORRENTS, KINOZAL, NNMCLUB, RUTRACKER).
- 2 indexers configured (kinozal, rutracker).
- 1 skipped â€” NNMCLUB (no Jackett indexer in 620-entry catalog; native plugin handles it).
- 1 errored â€” IPTORRENTS (Jackettâ€™s iptorrents indexer requires a cookie, env only had username/password).

Three end-user gaps remain:

1. **No in-dashboard visibility** of which Jackett indexers are configured, their status, or the autoconfig run history. Users must visit Jackettâ€™s UI on :9117 for any introspection or change.
2. **No in-dashboard management** â€” adding, editing, removing indexers, browsing the 620-indexer catalog, or fixing credential issues (like the IPTORRENTS_COOKIES gap) all require leaving Dâˆƒ4D±D°.
3. **No persistent canonical store** for credentials, autoconfig history, or catalog metadata. State today is split across .env (plaintext, single-process), per-indexer JSON files in config/jackett/, and in-memory results.

2. Goals

- A Dâˆƒ4D±D°-native UI for full Jackett management (browse 620-indexer catalog, add/edit/remove, test, browse autoconfig history).
- A SQLite system database (config/boba.db) as the persistent canonical store for credentials (encrypted), indexer state, catalog cache, autoconfig run history, and indexer-name overrides.

- Credentials encrypted at rest with AES-256-GCM using BOBA_MASTER_KEY (auto-generated into .env on first boot if missing).
- Bidirectional sync: .env â†’ DB on first run; UI changes write to BOTH DB (encrypted) AND .env (plaintext mirror) AND Jackett (live config).
- Strong leak-prevention tests asserting no credential value ever appears in logs or HTTP responses.
- Begin Go-canonical migration with a self-contained subsystem (Jackett UI), without disrupting the Python merge service that serves search.
- Zero loose ends at done-time: explicit Definition-of-Done checklist (Â§10) drives the implementation plan.

- Porting the merge search path (`download-proxy/src/merge_service/search.py` and friends) to Go. Stays Python until Phase 2.
- Porting hooks / schedules / theme / downloads endpoints to Go. Stays Python.
- Replacing nova3 search plugins (`plugins/*.py`) with Go equivalents. They run inside qBittorrent and must remain Python.
- A new login system for the dashboard. Re-uses existing `admin/admin` session auth.
- Postgres/MySQL support. SQLite only.
- Multi-user authorization. Single-tenant home-server use.

4.1 Process layout

The Angular app calls each backend on its own port; no reverse proxy is added. The existing Python `/api/v1/jackett/autoconfig/last` endpoint is removed; the Go service owns the entire `/api/v1/jackett/*` namespace.

4.2 Container layout

`docker-compose.yml` gains one service:

```
boba-jackett:
  build:
    context: ./qBitTorrent-go
    dockerfile: Dockerfile.jackett          # new, separate from existing Do
  container_name: boba-jackett
  environment:
    - JACKETT_URL=http://localhost:9117
    - JACKETT_API_KEY=${JACKETT_API_KEY:-}
    - BOBA_MASTER_KEY=${BOBA_MASTER_KEY:-}
    - BOBA_DB_PATH=/config/boba.db
    - BOBA_ENV_PATH=/host-env/.env
    - PORT=7189
    - LOG_LEVEL=INFO
  volumes:
    - ./config:/config                      # boba.db lives here
    - ./config/jackett:/jackett-config:ro
    - ./env:/host-env/.env                 # bind-mount for atomic updates
  network_mode: host
  restart: unless-stopped
  mem_limit: 256m
  pids_limit: 256
  oom_score_adj: 500
  depends_on:
    jackett:
      condition: service_healthy
  healthcheck:
    test: ["CMD-SHELL", "wget -qO- http://localhost:7189/healthz || exit 1"]
    interval: 30s
    timeout: 5s
    retries: 5
    start_period: 30s
```

`start.sh` extends `extract_jackett_key` with a similar `ensure_boba_master_key` step that runs *before* `start_container`, generating + writing `BOBA_MASTER_KEY` to `.env` if missing. The Go binary itself ALSO performs this generation at startup (defense in depth and also for cases where the host script is bypassed).

4.3 Go module layout

```
qBitTorrent-go/
├── cmd/
│   ├── qbittorrent-proxy/main.go      # existing
│   ├── webui-bridge/main.go          # existing
│   └── boba-jackett/main.go           # NEW
└── internal/
```

```

â",      â"œâ"€â"€ api/                # existing handlers stay
â",      â"œâ"€â"€ client/              # existing
â",      â"œâ"€â"€ jackett/            # NEW â€" autoconfig port +
â",      â",      â"œâ"€â"€ autoconfig.go
â",      â",      â"œâ"€â"€ catalog.go
â",      â",      â"œâ"€â"€ client.go
â",      â",      â"œâ"€â"€ matcher.go
â",      â",      â""â"€â"€ *_test.go
â",      â"œâ"€â"€ db/                  # NEW â€" SQLite + migratio
â",      â",      â"œâ"€â"€ conn.go
â",      â",      â"œâ"€â"€ crypto.go      # AES-256-GCM helpers
â",      â",      â"œâ"€â"€ migrate.go     # embedded migrations ste
â",      â",      â"œâ"€â"€ migrations/
â",      â",      â",      â""â"€â"€ 001_init.sql
â",      â",      â""â"€â"€ repos/        # one repo per table
â",      â",      â"œâ"€â"€ credentials.go
â",      â",      â"œâ"€â"€ indexers.go
â",      â",      â"œâ"€â"€ catalog.go
â",      â",      â"œâ"€â"€ runs.go
â",      â",      â""â"€â"€ overrides.go
â",      â"œâ"€â"€ envfile/              # NEW â€" atomic .env reade
â",      â",      â"œâ"€â"€ parse.go
â",      â",      â"œâ"€â"€ write.go        # tmp + fsync + rename
â",      â",      â""â"€â"€ *_test.go
â",      â"œâ"€â"€ jackettapi/           # NEW â€" HTTP handlers for
â",      â",      â"œâ"€â"€ credentials.go
â",      â",      â"œâ"€â"€ indexers.go
â",      â",      â"œâ"€â"€ catalog.go
â",      â",      â"œâ"€â"€ runs.go
â",      â",      â"œâ"€â"€ overrides.go
â",      â",      â"œâ"€â"€ auth_middleware.go # shared admin/admin sess
â",      â",      â""â"€â"€ *_test.go
â",      â"œâ"€â"€ logging/              # NEW â€" credential-redact
â",      â",      â"œâ"€â"€ redactor.go
â",      â",      â""â"€â"€ *_test.go
â",      â""â"€â"€ service/              # existing
â""â"€â"€ Dockerfile.jackett           # NEW

```

All shared mutable state across goroutines uses `digital.vasic.concurrency/pkg/safe` (CONST-13). No bare `sync.Mutex` + `map` patterns.

5. SQLite Schema (config/boba.db)

Migration `001_init.sql`:

Note: PRAGMA settings (`journal_mode=WAL`, `foreign_keys=ON`, `synchronous=NORMAL`) live in the connection DSN in `internal/db/conn.go`, not in this migration file. SQLite rejects PRAGMA `synchronous` inside a transaction, and the migrations stepper wraps each migration in a transaction for atomic apply. Setting them per-connection is the only place they actually take effect for `journal_mode` and `foreign_keys` anyway.

```

CREATE TABLE credentials (
  name          TEXT PRIMARY KEY,
  kind          TEXT NOT NULL CHECK (kind IN ('userpass', 'cookie')),
  username_enc  BLOB,
  password_enc  BLOB,
  cookies_enc   BLOB,
  created_at    DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at    DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  last_used_at  DATETIME
);

CREATE TABLE indexers (
  id            TEXT PRIMARY KEY,
  display_name  TEXT NOT NULL,
  type         TEXT NOT NULL,
  configured_at_jackett INTEGER NOT NULL,
  linked_credential_name TEXT REFERENCES credentials(name) ON DELETE SET NULL,
  enabled_for_search INTEGER NOT NULL DEFAULT 1,
  last_jackett_sync_at DATETIME,
  last_test_status TEXT,
  last_test_at DATETIME
);
CREATE INDEX idx_indexers_linked_cred ON indexers(linked_credential_name);

CREATE TABLE catalog_cache (
  id            TEXT PRIMARY KEY,
  display_name  TEXT NOT NULL,
  type         TEXT NOT NULL,
  language     TEXT,
  description   TEXT,
  template_fields_json TEXT NOT NULL,
  cached_at    DATETIME NOT NULL
);
CREATE INDEX idx_catalog_cached_at ON catalog_cache(cached_at);

CREATE TABLE autoconfig_runs (
  id            INTEGER PRIMARY KEY AUTOINCREMENT,
  ran_at        DATETIME NOT NULL,
  discovered_count INTEGER NOT NULL,
  configured_now_count INTEGER NOT NULL,
  errors_json   TEXT NOT NULL DEFAULT '[]',
  result_summary_json TEXT NOT NULL
);
CREATE INDEX idx_runs_ran_at ON autoconfig_runs(ran_at DESC);

CREATE TABLE indexer_map_overrides (
  env_name      TEXT PRIMARY KEY,
  indexer_id    TEXT NOT NULL,
  created_at    DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);

-- IF NOT EXISTS + INSERT OR IGNORE because Migrate() bootstraps this
-- same table first so it can query for applied versions before running
-- the migration body. Idempotency keeps fresh and resumed boots safe.

```

```
CREATE TABLE IF NOT EXISTS schema_migrations (
  version      INTEGER PRIMARY KEY,
  applied_at   DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);

INSERT OR IGNORE INTO schema_migrations (version) VALUES (1);
```

5.1 Encryption envelope

*_enc columns store nonce(12B) || ciphertext_with_GCM_tag. Helper internal/db/crypto.go:

```
func Encrypt(key []byte, plaintext string) ([]byte, error) // returns non
func Decrypt(key []byte, blob []byte) (string, error)
```

Empty values are stored as NULL, never as encrypted-empty (avoids leaking presence vs. absence).

5.2 File permissions

Strictly enforced at startup; failure aborts boot:

- config/boba.db and config/boba.db-wal, config/boba.db-shm â†’ 0600
- .env â†’ 0600
- A challenge script verifies these in the running container.

6. Bootstrap & Migration Flow

On every boba-jackett start:

1. **Load** .env from BOBA_ENV_PATH. If file missing, abort with clear error.
2. **Ensure** BOBA_MASTER_KEY:
 - If absent or empty in .env, generate 32 bytes via crypto/rand, hex-encode (64 chars).
 - Append to .env atomically with this header block:


```
# === BOBA SYSTEM ===
# Master key for credential encryption-at-rest in config/boba.db.
# DO NOT LOSE THIS " credentials become unrecoverable without it
# To rotate: see docs/BOBA_DATABASE.md Â§ "Key Rotation".
BOBA_MASTER_KEY=<hex>
```
 - Re-read .env. Log "BOBA_MASTER_KEY auto-generated and persisted to .env".
3. **Open DB** at BOBA_DB_PATH. If file missing, create + apply all migrations in internal/db/migrations/.
4. **Run any pending migrations** (compare schema_migrations.version to embedded migrations). Each migration runs in its own transaction.

5. **First-run import** (only when `credentials` table is empty):
 - Scan loaded env for `<NAME>_USERNAME/_PASSWORD/_COOKIES` triples (same regex as autoconfig: `^([A-Z][A-Z0-9_]+?)_(USERNAME|PASSWORD|COOKIES)$`).
 - Apply the same denylist as the existing autoconfig (QBITTORRENT, JACKETT, WEBUI, PROXY, MERGE, BRIDGE).
 - For each complete bundle, insert encrypted row.
 - Log "imported N credentials from .env".
6. **Idempotent re-import** (`credentials` table non-empty):
 - Insert ONLY env triples whose name is not already a row. **Existing rows are not overwritten** – DB wins.
 - Log "imported N new credentials from .env, M already in DB".
7. **Apply credential map overrides**: load `indexer_map_overrides` rows; merge with `JACKETT_INDEXER_MAP` env var (DB wins).
8. **Run autoconfig once**: same `autoconfigure_jackett` semantics as the Python impl, but reading creds from DB and overrides from DB. Result row inserted into `autoconfig_runs`.
9. Start HTTP server on PORT.

7. Sync Semantics

Trigger	DB write	.env write	Jackett write
User adds/edits cred via dashboard	âœ… encrypted upsert	âœ… atomic mirror (tmp+fsync+rename)	âœ… replay autoconfig for that single tracker
User removes cred via dashboard	âœ… row delete	âœ… remove <code><NAME>_*</code> triple	âœ… DELETE /api/v2.0/indexers/{id}
User edits .env manually + restart	âœ… insert if not exists (step 6)	(already done)	âœ… autoconfig on boot
User adds indexer via Jackett UI directly (:9117)	âœ… upsert <code>indexers</code> row on next sync poll	âœ… no env write	(already done by user)
User toggles <code>enabled_for_search</code>	âœ… update row	âœ…	âœ… (search filter applied at query time)
Catalog refresh (TTL 24h or manual button)	âœ… <code>catalog_cache</code> upsert	âœ…	âœ… (read-only fetch)
BOBA_MASTER_KEY regenerated (manual rotation)	âœ… re-encrypt all rows in transaction	âœ… overwrite key	âœ…

7.1 Atomic .env writes

Implemented in `internal/envfile/write.go`:

1. Read entire current .env into memory.
2. Apply requested mutation (insert/update/delete one or more triples).
3. Write to `<envpath>.tmp` with mode `0600`.

4. `fsync(.tmp).`
5. `rename(.tmp, <envpath>)` (atomic on POSIX same-FS).
6. `fsync(parent_dir).`

A single `sync.Mutex` in the writer guards against concurrent dashboard requests.

7.2 Conflict policy

Dashboard writes always win. If a user manually edits `.env` while the service is running and then triggers a restart, the manual change is imported only if the row isn't already in the DB. The DB is the running source of truth.

For users who NEED a manual `.env` edit to take effect on a tracker that already has a DB row: documented procedure is "delete the row via dashboard, then restart" (or `POST /api/v1/jackett/credentials/{name}` overwrite via API).

8. API Surface (Go service on :7189)

All endpoints under `/api/v1/jackett/`. All POST/PATCH/DELETE require the existing admin/admin session cookie (validated via shared middleware `internal/jackettapi/auth_middleware.go`).

8.1 Credentials

```
GET    /credentials
      200 [{name, kind, has_username, has_password, has_cookies,
           created_at, updated_at, last_used_at}]
      Never returns plaintext values.

POST   /credentials
      Body: {name, username?, password?, cookies?}
      PATCH semantics "only fields present are updated."
      200 {name, kind, has_username, ...}

DELETE /credentials/{name}
      Cascade: linked_credential_name 200 NULL on indexers; ON DELETE SET
      204
```

8.2 Indexers

```
GET    /indexers
      200 [{id, display_name, type, configured_at_jackett,
           linked_credential_name, enabled_for_search,
           last_jackett_sync_at, last_test_status, last_test_at}]

POST   /indexers/{id}
      Body: {credential_name, extra_fields?}
      Configures (or reconfigures) the indexer in Jackett. Upserts row.
      200 indexer object

DELETE /indexers/{id}
      Removes from Jackett + deletes row.
```


â†’ 204

POST /indexers/{id}/test
Triggers a test query against the indexer.
â†’ 200 {status: "ok"|"auth_failed"|"unreachable"|"empty_results",
Side-effect: updates last_test_status + last_test_at.

PATCH /indexers/{id}
Body: {enabled_for_search: bool}
â†’ 200 indexer object

8.3 Catalog

GET /catalog?search=&type=&language=&page=&page_size=
Paginated query over catalog_cache. page_size defaults 50, max 200.
â†’ 200 {total, page, page_size, items: [{id, display_name, type,
language, description,
required_fields: [...]}]}

POST /catalog/refresh
Force re-fetch from Jackett. Returns when complete (typically <2s).
â†’ 200 {refreshed_count, errors: []}

8.4 Autoconfig runs

GET /autoconfig/runs?limit=50
â†’ 200 [{id, ran_at, discovered_count, configured_now_count,
error_count}]
Paginated by limit only (no offset; UI shows last N).

GET /autoconfig/runs/{id}
â†’ 200 full redacted result_summary_json

POST /autoconfig/run
Triggers a synchronous autoconfig run.
â†’ 200 full redacted run summary

8.5 Indexer-name overrides

GET /overrides
â†’ 200 [{env_name, indexer_id, created_at}]

POST /overrides
Body: {env_name, indexer_id}
Upsert.
â†’ 200 row

DELETE /overrides/{env_name}
â†’ 204

8.6 Health

```
GET /healthz
  200 {status, db_ok, jackett_ok, version, uptime_s}
```

8.7 OpenAPI

GET /openapi.json exposes a generated OpenAPI 3.1 spec; contract tests (Layer 6) validate live responses against it.

9. Dashboard UI (Angular)

Two new top-level routes under /jackett. Implementation uses Angular Material to match the existing dashboard.

9.1 /jackett/credentials

Table view columns: name, kind badge, presence badges (has-user / has-pass / has-cookies), last-used-at, actions (edit / delete).

“Add credential” button modal with fields: - Tracker name (uppercase, validated against `^[A-Z][A-Z0-9_]+$`) - Kind selector (userpass / cookie) - Conditional fields based on kind

Edit modal masked input fields. **Empty field = unchanged** (PATCH semantics, mirrored from API). User may explicitly clear a field via a “Clear” button next to it.

Confirmation modal on delete shows linked indexers that will lose their credential link.

9.2 /jackett/indexers

Three tabs:

Tab 1: Configured

Table of currently configured indexers (`indexers.configured_at_jackett = 1`). Columns: status badge (green=ok, yellow=stale, red=auth_failed/unreachable), id, display name, linked credential, enabled-for-search toggle, last test, actions (test / remove / edit).

Tab 2: Browse Catalog

Searchable table over catalog cache. Filters: text search (name + id + description), type (public/private/semi-private), language. Paginated 50/page.

Each row has an “Add” button. Clicking opens a configuration modal pre-populated with: - The indexer’s required fields (from `template_fields.json`) - A credential picker auto-suggests a credential whose name fuzzy-matches the indexer (the same matcher as autoconfig), or “Create new credential” inline - A “Save” button that configures via Jackett + writes the row

Tab 3: Autoconfig History

Last 50 run entries (most recent first). Click row `â†’` expands to show the full redacted `result_summary_json` with syntax-highlighted JSON viewer. `â€œRun autoconfig nowâ€` button at the top.

9.3 IPTorrents cookie capture flow

When the user clicks `â€œAddâ€` on `iptorrents` (or any indexer whose required-fields include `cookie / cookies / cookieheader`), the modal opens with a step-by-step instruction panel:

1. Open `https://iptorrents.com` in a new browser tab and log in.
2. Open DevTools (F12) `â†’ Application â†’ Storage â†’ Cookies â†’ iptorrents.com`.
3. Copy the values of the `uid` and `pass` cookies.
4. Format as `uid=<value>; pass=<value>` and paste below.

Single textarea for the cookie blob, `â€œSaveâ€` writes encrypted cookie row + triggers configure for that indexer.

9.4 NNMClub clarification

A small info banner on `/jackett/credentials` for any tracker name that exists in the credentials table but has no Jackett indexer match (after catalog has been loaded once):

NNMCLUB credentials present but no Jackett indexer matches. NNMClub is served by the native nova3 plugin (`plugins/nnmclub.py`) `â€”` these credentials are still used. No further action required.

The autoconfig result `skipped_no_match` field gets a parallel `served_by_native_plugin` field for any name that matches a known native-plugin filename in `plugins/`.

9.5 Routing + nav

Top nav gains a single `â€œJackettâ€` entry. Click `â†’ /jackett` (default child route `/jackett/credentials`).

10. Test Strategy

Resource-limited per CONST-09: `GOMAXPROCS=2 nice -n 19 ionice -c 3` for all `go test` invocations. Container memory limits already in compose.

10.1 Layer 1 `â€”` Unit (`Go_test.go`, -short, mocks OK)

Module	Tests
<code>internal/db/crypto</code>	round-trip; nonce uniqueness over 100k samples; tamper detection (flip 1 bit <code>â†’</code> fail); empty plaintext rejected
<code>internal/db/migrate</code>	

Module	Tests
	apply 0â†’latest; idempotency; transaction rollback on bad SQL; embedded files match disk
internal/envfile	parse all forms (quoted/unquoted/comments/blanks); atomic-write under crash sim (kill -9 between tmp + rename); whitespace + duplicate-key handling
internal/jackett/matcher	port of Python tests verbatim â€” fuzzy threshold, override precedence, ambiguous detection
internal/jackett/autoconfig	port of Python tests â€” denylist, complete/incomplete bundles, total-timeout, error envelope
internal/logging/redactor	every credential value present in the database is replaced by *** in any log line

10.2 Layer 2 â€” Integration (real SQLite + real .env, no Jackett)

tests/integration/jackett_db_test.go:

- Bootstrap from empty .env â†’ master key generated, written, file mode 0600.
- Bootstrap from .env with N triples â†’ all imported, encrypted, table count = N.
- Restart with same .env â†’ no re-imports, log shows â€œ0 new, N already in DBâ€œ.
- UI add cred â†’ DB row + .env line both present and consistent.
- UI delete cred â†’ DB row gone + .env line gone.
- Concurrent dashboard writes via 50 goroutines â†’ no .env corruption (line count + parse check after each).

10.3 Layer 3 â€” E2E (real Go service + real Jackett container + real .env)

tests/e2e/jackett_management_test.go:

- Boot full stack; add RUTRACKER cred via API â†’ autoconfig runs â†’ indexer appears in Jackettâ€™s /api/v2.0/indexers.
- Browse catalog â†’ 620 entries returned.
- Add iptorrents with cookie via API â†’ indexer configured in Jackett with cookie value.
- Test indexer â†’ status=â€œokâ€œ returned.
- Toggle enabled_for_search â†’ search request to merge service excludes that indexer.
- Delete cred â†’ indexer gone from Jackett.
- Pull JACKETT_INDEXER_MAP from env â†’ DB override row created â†’ autoconfig uses it.

10.4 Layer 4 â€” Security / Penetration (THE leak-prevention requirement)

tests/security/credential_leak_test.go + challenges/scripts/credential_leak_grep_challenge.sh:

- Insert 6 unique high-entropy credentials (so any leak is greppable with low false-positive rate).

- Drive a full E2E flow exercising every endpoint in Â§8.
- Capture: all log output (stdout + stderr + file), every HTTP response body across the flow, full boba.db hex dump, full process /proc/<pid>/environ.
- Grep all captures for the literal credential values.
- **Assertion:** 0 hits in logs, 0 hits in HTTP responses, 0 hits in /proc/<pid>/environ (env vars containing the values must not be set on the process â€” only BOBA_MASTER_KEY should be).
- Hits in boba.db ARE allowed but must be encrypted (verify by attempting to decrypt with the wrong key â†’ fail; with the right key â†’ match).

Static scans: - gosec ./... clean - bandit -r download-proxy/src/ clean

File permissions: - config/boba.db, config/boba.db-wal, config/boba.db-shm mode 0600 - .env mode 0600 - A challenge script asserts these in the running container.

10.5 Layer 5 â€” Benchmark

internal/db/repos/*_bench_test.go:

- BenchmarkCatalogQuery â€” over the 620-row cache with these scenarios: (a) no filter (full page), (b) text search 3-char prefix, (c) type=private filter alone, (d) text search + language filter combined; each scenario p99 < 50ms on host hardware.
- BenchmarkEncryptDecrypt â€” p99 < 1ms per op.
- BenchmarkAutoconfigFullRun â€” full autoconfig with 5 mock indexers; p99 < 5s.

10.6 Layer 6 â€” Contract

OpenAPI 3.1 generated at build time. tests/contract/openapi_test.go hits every endpoint and validates the response shape against the spec.

10.7 Layer 7 â€” Challenge (CONST-02 + CONST-32 reproduction-before-fix)

In challenges/scripts/:

1. cred_roundtrip_challenge.sh â€” POST cred â†’ GET should return metadata only â†’ DB row encrypted â†’ .env line plaintext.
2. env_db_drift_challenge.sh â€” make 100 changes via API, assert .env and DB are byte-equivalent in semantics after each.
3. iptorrents_cookie_flow_challenge.sh â€” follow the documented flow â†’ indexer configured with cookie field populated.
4. master_key_autogen_challenge.sh â€” fresh .env without BOBA_MASTER_KEY â†’ service starts â†’ key appears in .env with the documented header â†’ restart â†’ key is unchanged (idempotent).
5. credential_leak_grep_challenge.sh â€” wrapper around the Layer 4 test for inclusion in run_all_challenges.sh.
6. nnmclub_native_plugin_clarification_challenge.sh â€” autoconfig output for NNMCLUB shows served_by_native_plugin flag.
7. boba_db_file_perms_challenge.sh â€” assert 0600 on all DB and env files.

All registered in challenges/scripts/run_all_challenges.sh.

11. Definition of Done (the “no loose ends” requirement)

A single checklist that MUST be “... before declaring this work complete:

1. **All endpoints in Â§8 implemented** + Layer 2 integration tested + Layer 3 E2E tested against live stack.
2. **Both UI pages implemented** + manually verified in browser via Playwright walkthroughs (golden path + 5 edge cases per page documented in implementation plan).
3. **All 7 test layers green:**
 - Layer 1 (unit) `go test -short -race ./...` “pass
 - Layer 2 (integration) “pass
 - Layer 3 (E2E) “pass
 - Layer 4 (security) “leak grep 0 hits; gosec/bandit clean; file perms enforced
 - Layer 5 (bench) “all p99 thresholds met
 - Layer 6 (contract) “OpenAPI matches every response
 - Layer 7 (challenges) “all 7 challenge scripts pass via `run_all_challenges.sh`
4. **Plan doc reconciled:** `docs/superpowers/plans/2026-04-26-jackett-autoconfig-clean-rebuild.md` checkboxes updated to reflect what shipped (currently 84 unchecked, 0 checked even though Layers 1-7 are in commits).
5. **Documentation updated:**
 - `CLAUDE.md` Â§ “Architecture” “adds boba-jackett service + ports map
 - `AGENTS.md` “same updates
 - `docs/JACKETT_INTEGRATION.md` “auto-config section rewritten as DB-backed
 - `docs/BOBA_DATABASE.md` “NEW: schema reference, master key lifecycle, key rotation procedure, backup procedure, recovery from lost key
6. **Bugfix doc:** every bug surfaced + fixed during implementation logged in `docs/issues/fixed/BUGFIXES.md` per CONST-10.
7. `IPTORRENTS_COOKIES` documented in `docs/JACKETT_INTEGRATION.md` Â§ “Cookie-only indexers” and surfaced in the IPTorrents add modal.
8. **NNMClub clarification** in autoconfig output AND in dashboard banner.
9. **Pre-test commit + push to all upstreams** “every change committed via Conventional Commits and pushed to **every configured remote of the main repo** (currently `origin`, `github`, `upstream` “all aliased to `git@github.com:milos85vasic/qBittorrent.git`, but each must receive its own `git push <remote> <branch>` invocation) **before** containers are booted for the final test/challenge run. If submodules are added in the future (none today “verified `.gitmodules` absent on 2026-04-27), each must also be committed + pushed to all its remotes before parent-repo push.
10. **Final demo block** in PR body with pasted output from the real end-to-end run (CONST Definition of Done).
11. **Open-issues sweep:** `grep -rE 'TODO|FIXME|XXX|HACK' qBittorrent-go/download-proxy/src/ frontend/src/ docs/` produces zero new entries from this work. Pre-existing entries are left alone unless touched.
12. **CONST-013 audit:** no bare `sync.Mutex` + `map/slice` in any new Go code; all mutable shared collections use `safe.Store/safe.Slice`.
13. **CONST-033 verification:** `bash challenges/scripts/no_suspend_calls_challenge.sh` AND `bash challenges/scripts/host_no_auto_suspend_challenge.sh` both pass after all changes are committed.

12. Risks & Mitigations

Risk	Mitigation
Master key lost → all credentials unrecoverable	(a) Auto-generated key written to .env with prominent header; (b) docs / BOBA_DATABASE.md documents backup procedure; (c) on key-mismatch detection at startup, abort with explicit “key mismatch” restore your .env from backup, or wipe boba.db to re-import from .env → error tmp + fsync + rename + parent-dir fsync; concurrent-writer mutex; integration test simulates kill -9 mid-write
.env corruption during atomic write	OpenAPI spec is authoritative; Python / api / v1 / jackett / autoconfig / last endpoint is REMOVED in this work → single owner
Two backends drift on shared concepts (autoconfig output shape)	Server-side pagination + index on cached_at; benchmark gate at p99 < 50ms
Browse-catalog page slow with 620 rows	Documented in docs / JACKETT_INTEGRATION.md; future dashboard banner on detected drift
Manual .env edits silently ignored after first import	Run inside a single SQLite transaction; on failure roll back, keep old key
Re-encryption migration breaks if interrupted	

13. Out of Scope (Phase 2+)

- Port merge search to Go.
- Multi-user auth / per-user credential isolation.
- Postgres backend.
- Master key rotation UI (manual procedure documented; scripted in Phase 2).
- OAuth-based credential acquisition for trackers that support it.
- Backup/restore UI (manual cp documented).

14. References

- docs / JACKETT_INTEGRATION.md → current Jackett integration documentation
- docs / superpowers / specs / 2026-04-26-jackett-autoconfig-clean-rebuild-design.md → predecessor spec for the autoconfig itself
- docs / superpowers / plans / 2026-04-26-jackett-autoconfig-clean-rebuild.md → implementation plan for the autoconfig (checkboxes need reconciliation per §11.4)
- download-proxy/src/merge_service/jackett_autoconfig.py → reference Python implementation to port
- CLAUDE.md → “Universal Mandatory Constraints” → CONST-01, CONST-02, CONST-09, CONST-10, CONST-11, CONST-13, CONST-32, CONST-33
- CONSTITUTION.md → CONST-033 → host power management ban